

Standard Operating Procedure

Card Reader Installation & Wiegand Wiring

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Owner	Apex – Operations
Approved By	_____

1. Purpose

This procedure establishes standard practices for mounting, wiring, and testing card readers on a Kantech access control system, using the Wiegand interface.

2. Scope

Applies to all Apex technicians installing card/credential readers (proximity, smart card, or keypad/prox combination) wired via the Wiegand protocol to a Kantech controller.

3. Wiegand Cable Specification

- Use dedicated, purpose-built Wiegand reader cable where possible — typically 5-conductor (or more), 22 AWG or 18 AWG, overall shielded, non-twisted for the data pair. Manufacturers publish this as the reference standard for their readers.
- Do not use standard twisted-pair Cat5e/Cat6 cable for Wiegand runs if avoidable — the twist on the D0/D1 data conductors can introduce timing distortion. If Cat5e/Cat6 must be used, ensure D0 and D1 are NOT run on the same twisted pair with each other, and reinforce grounding.
- Run power and data conductors together in the same jacket (standard for reader cable), but keep the overall cable run separated from line-voltage power wiring and noisy loads (ballasts, motors, VFDs) — do not share conduit.

4. Maximum Cable Run

- 18 AWG, shielded, good-quality cable: approximately 500 ft (150 m) maximum, in ideal conditions.
- 22 AWG: maximum run drops significantly, to roughly 250 ft under the same ideal conditions.
- These are theoretical/ideal-condition maximums — always confirm the specific reader manufacturer's published maximum distance, since it varies by model and required operating current. Environmental noise, cable quality, and voltage drop under the reader's actual current draw all reduce real-world achievable distance below the theoretical number.
- Where a run approaches the practical limit, consider a heavier gauge (18 AWG over 22 AWG) or relocating the controller/expansion point closer to the reader rather than pushing a marginal run into service.

5. Grounding

- Ground the cable's drain/shield wire at one end only — typically at the controller/panel end. Grounding at both ends creates a ground loop and can introduce noise onto the data lines.
- Common the reader's power supply ground to the controller ground where the reader uses a separate power supply (required if reader current draw exceeds the controller's rated output, commonly around 150 mA — confirm against the specific controller and reader).

6. Reader Mounting

- Mount at a consistent height across the site for a professional, ADA-considerate installation — confirm the site's standard mounting height before starting, rather than defaulting to an arbitrary height per door.

- Where a reader is mounted outdoors or in a wet location, confirm the reader's environmental rating (IP rating) is suitable and use a weatherproof back box/gasket per the manufacturer's instructions.
- Avoid mounting a reader directly on or immediately adjacent to metal door frames without a standoff/spacer where the manufacturer specifies one — proximity readers in particular can have reduced read range when mounted directly on metal.

7. Wiring at the Controller

- Confirm which Wiegand input the reader is being wired to, and confirm the controller's door/reader assignment matches the physical door before commissioning — a reader wired to the wrong logical input is a common and easily-avoided commissioning error.
- Do not power the reader from the wrong voltage tap — confirm the reader's rated voltage (commonly 12VDC, sometimes 5VDC for some legacy readers) against what the controller/power supply provides at that terminal before connecting.
- Where the controller provides reader LED and buzzer control outputs, wire them if the application calls for visual/audible feedback — confirm polarity and current rating against the reader's requirements.

8. Testing

- Present a known-good credential and confirm a clean, consistent read — test multiple times, not just once, since a marginal cable run can produce intermittent rather than consistent failures.
- Confirm the correct door/reader is reporting the credential read in the access control software — verifying the logical assignment matches the physical door.
- Test reader LED/buzzer feedback if wired, confirming correct behavior for both valid and invalid credential presentations.
- Where a run is long or borderline, test under realistic conditions (not just a quick bench test) — confirm reliable reads over multiple attempts at the actual installed distance.
- Record reader location, cable run length, and gauge on the site door schedule/configuration sheet.

9. References

- Manufacturer installation instructions for the specific reader model in use (govern where more restrictive than this SOP)
- Wiegand interface specification (industry de facto standard)
- SOP-LV-010 — KT-1/KT-2/KT-4/KT-400 Controller Default & Configuration
- SOP-LV-013 — RS-485 Bus Design & Installation